

Stacked Polyphase Bridge Converters for High-Voltage Electric Traction Drives: A Review of Topology, Modulation, Control, Semiconductor Scaling, and Research Challenges

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Abstract—Higher dc-link voltage is increasingly attractive in electric traction, but it raises semiconductor blocking-voltage, switching-transient, insulation, and electromagnetic-compatibility demands. The stacked polyphase bridge (SPB) addresses the blocking-voltage problem by connecting three-phase two-level bridges in series on the dc side, each supplying an electrically separated winding group — a voltage-partitioning approach that accesses lower-voltage semiconductor classes and supports modular converter-machine integration, at the cost of added winding, dc-link, sensing, gate-drive, and insulation-coordination overhead. Although SPB has been studied for over a decade, the literature remains scattered across topology, modulation, common-mode voltage, dc-link balancing, multiphase-machine design, semiconductor technology, and fault-tolerant control, without a common system-boundary evaluation. This review consolidates SPB’s evolution, operating principle, and machine-integration options (aligned, spatially shifted, and physically segmented winding groups); derives the double Fourier series basis for carrier-interleaved dc-link ripple and its closed-form estimate, and shows that the same carrier-synchronization freedom can instead cancel switching-induced common-mode voltage, illustrated for a two-cell stack; analyzes dc-link voltage-balancing stability from the exact nonlinear cell model across centralized, common-duty-ratio, and distributed control; and examines semiconductor scaling, commercial-device screening, and fault-tolerant reconfiguration. A cell-count-to-voltage-class mapping developed for a 1200-V dc link shows that lower device-voltage classes can yield nonlinear semiconductor-level gains while converter and machine hardware scale roughly with cell count, so the preferred cell count is mission-dependent rather than fixed. Realizing this potential requires treating the battery/dc-link, semiconductor/converter, and machine sides as one co-design problem rather than as separate design axes – the direction this review identifies for future SPB research.

Index Terms—Electric vehicles, integrated motor drives, modular converters, multiphase machines, stacked polyphase bridge (SPB) converters.

I. INTRODUCTION

Electrification is pushing traction-drive propulsion and fast-charging power toward several hundred kilowatts, while system-level performance requirements continue to rise [1]. Increasing the dc-link voltage has emerged as a primary design lever for meeting both demands. On the propulsion side, a higher dc-link voltage improves efficiency and power density [2]. On the battery and charging side, it reduces

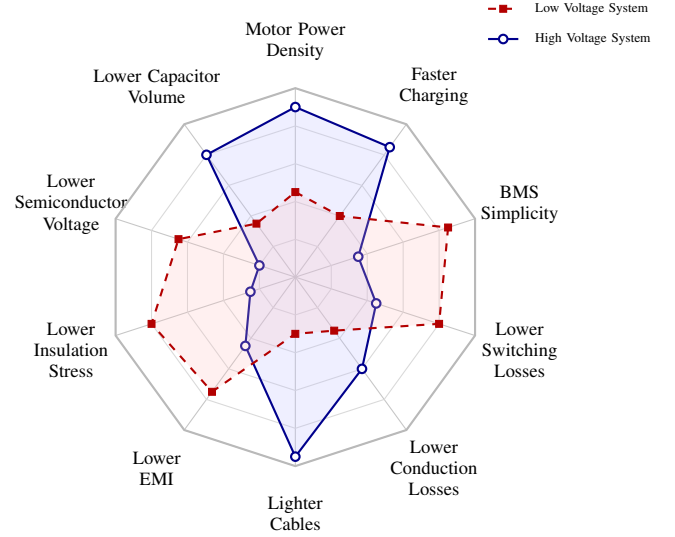


Fig. 1. Conceptual illustration of the qualitative system-level tradeoffs between a low-voltage and a high-voltage traction architecture.

the current required for a given power level, lowering cable conduction loss and conductor cross-section while enabling higher charging power under a given current limit [3].

This trend, however, is not without consequence at the converter level. A higher dc-link voltage also brings more severe insulation-coordination and electromagnetic-compatibility (EMC) demands, together with added thermal and reliability stress on the power stage [2], [3]. Fig. 1 summarizes this qualitative tradeoff between a conventional low-voltage and a higher-voltage traction architecture. Raising the dc-link voltage therefore does not amount to a simple rescaling of an existing inverter design; it reshapes the architecture-level design space of the complete electric drive.

A. Literature and Industry Trends Toward Higher DC-Link Voltage

The trend motivated above is not merely conceptual: it is already established in academic prototypes, industrial inverter platforms, and production vehicles, as summarized in Table I. Reported academic demonstrators range from 400-V to 800-V and 1.2-kV converters, with still higher levels investigated for high-power passenger and heavy-duty applications [3], while industrial platforms increasingly target 800-V operation as a

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TABLE I

REPRESENTATIVE VOLTAGE AND POWER LEVELS IN PUBLISHED TRACTION-INVERTER STUDIES, INDUSTRIAL AND REFERENCE PLATFORMS, AND PRODUCTION BATTERY-ELECTRIC VEHICLES.

Published studies / demonstrators	Industrial / reference platforms	Production vehicles
Bertelshofer et al.: 400 V, 150 kW [6]	Bosch Gen. 4:400/800 V, 250/350 kW [7]	BMW i4 M50: 400 V, 400 kW [8]
Taha et al.: 800 V, 100 kW [9]	Wolfspeed reference design: 800 V, 300 kW [10]	Hyundai IONIQ 5 N: 800 V, 478 kW [11]
Al-Hmoud et al.: 1 kV, 200 kW [12]	Infineon eCAV platform: 800 V, 250 kW [13]	Volvo EX90 Performance: 800 V, 500 kW [14]
Wang et al.: 1.2 kV, 100 kW [15]	AVL PI850e: 800 V, 250 kW [16]	Porsche Taycan Turbo S: 800 V, 700 kW [17]
Ayaz et al.: 1.2 kV, 250 kW [18]	Punch IV5: 800 V, 600 kW [19]	Lucid Air Dream Performance: 900 V, 828 kW [20]

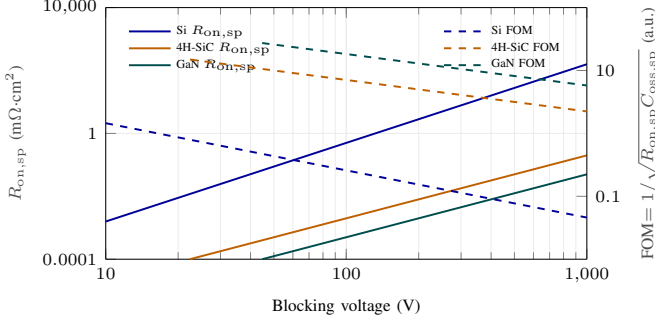


Fig. 2. Ideal one-dimensional unipolar specific on-state resistance $R_{on,sp}$ (left axis, solid) and the resulting high-frequency figure of merit $FOM = 1/\sqrt{R_{on,sp}C_{oss,sp}}$ (right axis, dashed, arbitrary units) versus blocking voltage for Si, 4H-SiC, and GaN.

baseline and production vehicles show a comparable shift from established 400-V architectures to a growing number of 800-V and higher-voltage platforms [4], [5]. This industry-wide direction, rather than an isolated research interest, motivates the semiconductor-level analysis that follows since the rising dc-link voltages translate directly into a semiconductor-level bottleneck.

B. Semiconductor Blocking-Voltage Tradeoff

In a conventional two-level voltage-source inverter, each active switch must block a voltage directly tied to the complete dc-link voltage. In unipolar power devices, a higher blocking-voltage rating imposes a fundamental tradeoff against both specific on-state resistance and switching performance, commonly captured by the figure of merit, as shown in Fig. 2 [21], [22]. Therefore, a lower-voltage device consistently achieves a higher FOM than a higher-voltage device of the same material.

Wide-bandgap (WBG) semiconductors shift this FOM favorably: SiC metal-oxide-semiconductor field-effect transistors (MOSFETs) now dominate high-voltage automotive drives, while GaN is particularly attractive at lower voltage owing to its fast switching behavior and superior FOM [21], [23]–[25]. Yet switching material does not remove the underlying coupling between dc-link voltage and device voltage class: irrespective of technology, a higher dc-link voltage still forces a higher-voltage, higher- $R_{on,sp}$ device and higher switching losses. Converter architectures that partition the dc-link voltage among multiple switching units, so each device blocks only a fraction of the total voltage, therefore become increasingly relevant as system voltage rises.

C. Multilevel Converters and Their Limitations

Multilevel converters constitute an established solution to this voltage-partitioning problem. NPC, T-type, flying-capacitor, and cascaded architectures all distribute the dc-link voltage among additional semiconductor positions or capacitor-voltage levels, reducing individual device voltage stress [26]–[29].

This voltage partitioning, however, comes at the cost of additional semiconductor devices, passive components, and gate drivers – each an additional potential failure point – without, by itself, yielding a modular converter-machine system: because the added semiconductor positions continue to feed a single, electrically common three-phase winding set, a fault at one position cannot generally be isolated and bypassed without disabling the drive. Fault tolerance, power redistribution, and post-fault reconfiguration can be implemented in conventional multilevel drives, but typically require additional redundancy, dedicated isolation paths, and specialized control layered on top of the base topology rather than following from it. A larger semiconductor count therefore does not yield a commensurate increase in fault tolerance, motivating an alternative form of voltage partitioning.

D. Stacked Polyphase Bridge

Unlike other multilevel converters, which distribute the dc-link voltage among additional semiconductor positions within a single converter, the stacked polyphase bridge (SPB) distributes the high-voltage conversion function among N repeated converter units. Each unit is a conventional three-phase two-level bridge connected in series on the dc side, supplying its own electrically separated three-phase winding group of the same machine, as illustrated in Fig. 3.

Under balanced operation, each cell's local dc-link voltage $V_{dc,k} \approx V_{dc}/N$, so the semiconductor voltage rating can be selected on the local cell voltage rather than the complete traction dc-link voltage – addressing the semiconductor bottleneck much as multilevel converters do. Thanks to each cell also drives its own separated winding group, SPB additionally has an inherently modular structure that multilevel converters lack. A faulted cell can, in principle, be isolated and bypassed while the remaining cells continue to drive the machine at reduced power. It is this combination of semiconductor voltage reduction and converter-machine modularity that distinguishes SPB from conventional multilevel and cascaded architectures and motivates its consideration as a promising candidate topology for next-generation traction drives.

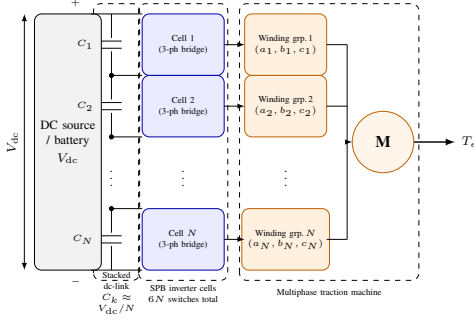


Fig. 3. Conceptual architecture of an N -cell SPB traction drive: series-connected local dc-link capacitors, series-connected three-phase two-level bridge cells, and their electrically separated machine winding groups.

Realizing this potential, however, introduces coupled design challenges in component count, dc-link dynamics, machine design, common-mode voltage, and fault handling, addressed throughout this review. These challenges have grown more tractable as neighboring research areas have matured: multiphase machines provide additional post-fault degrees of freedom, integrated modular motor drives show how repeated power-electronic units can be physically associated with individual machine sections, wide-bandgap semiconductors amplify performance differences among voltage classes, and distributed control coordinates modular converter units [30]–[35].

Despite more than a decade of SPB research, beginning with Norrga *et al.* [36], no study has yet brought these threads together to systematically assess its feasibility for electric-vehicle traction applications. The available literature remains distributed across converter topology, modulation, dc-link stability, multiphase-machine design, integrated motor drives, semiconductor technology, thermal management, distributed control, and fault-tolerant operation. Although these topics are strongly coupled in an SPB traction drive, they are rarely evaluated within a common system boundary, leaving a gap between a well-motivated architectural concept and a system-level feasibility assessment. Accordingly, this paper provides a system-level review of SPB traction-drive technology aimed at closing that gap.

The remainder of the paper is organized as follows. First, SPB’s historical development, operating principle, and modulation are consolidated. Second, the interaction between cell count and the electric machine is examined alongside multiphase-machine and integrated-modular-drive research, including the insulation-coordination overhead of series-connected cells. Third, common-mode voltage and dc-link dynamics and balancing are examined. Fourth, semiconductor voltage scaling and fault handling are treated as coupled architecture-level design problems. Finally, the remaining research gaps and automotive validation requirements are identified.

II. EVOLUTION OF THE STACKED POLYPHASE BRIDGE CONCEPT

SPB emerged from a broader development toward modular converter-machine systems. Earlier modular and integrated

modular motor drives (IMMDs) established building blocks that SPB later drew on – repeated converter-winding modules, machine segmentation, power sharing, and fault-tolerant operation [30], [37] – and series-connected IMMD modules showed that the total dc voltage could be shared among inverter segments to access lower-voltage devices [31], [38], [39]. Independently, other series-connected modular motor-drive topologies were proposed on the same dc-voltage-sharing principle [40], [41], and a KTH comparative study benchmarked SPB against other modular-converter candidates such as the parallel-connected polyphase bridges (PPBs) on machine design, losses, capacitor energy, cost, and redundancy [42], confirming SPB as one of several promising topology candidates.

The SPB topology itself was introduced in 2013 by Norrga *et al.* [36], followed by dedicated control and modulation work [43]. Related series-connected drives showed that machine-side degrees of freedom could regulate internal dc-link sharing [44], while Nikouie *et al.* formally established the dc-link stability problem and its balancing-controller requirements [45], [46], which identified dc-link balancing, rather than semiconductor voltage rating alone, as SPB’s first-order control challenge.

Development then shifted toward modulation, scalable control, and converter-machine co-design: modulation was shown to shape switching loss and dc-link harmonics [47], distributed control offered an alternative to centralized balancing [48], and interleaving and phase-count studies tied converter design to machine architecture [33], [49]–[51] – reframing SPB from a converter topology into a joint converter-machine design problem.

Most recently, work has moved from statically configured hardware toward actively reconfigurable systems: common-duty-ratio operation, decentralized balancing, and multi-agent power distribution have reduced reliance on centralized control [52]–[55], and online module isolation and reintroduction have demonstrated dynamic dc-bus reconfiguration that removes a faulted cell without interrupting torque production [56]. Modern GaN devices have in parallel renewed interest in SPB as a route to lower semiconductor voltage classes [57].

Fig. 4 places these and other representative milestones on a chronological timeline, together with the implication each result carries for the system-level review developed in this paper.

III. OPERATING PRINCIPLE AND TOPOLOGICAL CHARACTERISTICS

A. Basic Structure and Voltage Partitioning

An N -cell SPB consists of N conventional three-phase two-level voltage-source inverter cells connected in series on the dc side, with each bridge supplying one electrically separated three-phase winding group of the same machine [36], [43], [47]. The machine therefore presents N ac ports, while the converter cells share a single series dc path.

The dc-link voltage satisfies

$$V_{dc} = \sum_{k=1}^N V_{dc,k}, \quad V_{dc,k} \approx \frac{V_{dc}}{N} \quad (1)$$

- **2013** – Norrga *et al.* [36]: original SPB topology introduced
- **2014** – Jin *et al.* [43]: dedicated control and modulation methods established
- **2015** – Nikouie *et al.* [45], [46]: dc-link stability problem formally established
- **2017** – Jin *et al.* [47]: modulation shown to affect switching loss and dc-link harmonics
- **2017** – Zhang *et al.* [42]: SPB benchmarked against PPB-I/II and MHF converter candidates
- **2019** – Verkroost *et al.* [33], [49]–[51]: interleaving and phase-count tied to machine design
- **2023** – Mocevic *et al.* [52]: CDR balancing demonstrated
- **2024** – Verkroost *et al.* [53], [54], [56]: distributed balancing, power sharing, and online module reconfiguration
- **2024** – Rohner *et al.* [57]: 650-V GaN SPB vs. 3L flying-capacitor comparison

Fig. 4. Chronological milestones in SPB research, from the original topology through dc-link stability, modulation, distributed balancing, reconfiguration, and wide-bandgap integration.

under balanced operation. A basic implementation contains $6N$ active switches and N local dc-link capacitors. Increasing N reduces the nominal cell voltage, and hence the required semiconductor voltage class, but simultaneously increases the number of switches, gate drivers, sensors, capacitors, and machine-winding terminals that must be realized and managed. Fig. 5 shows this structure at switch level for two representative cells, making explicit the $6N$ -switch bridge realization, the local capacitors C_k , and the notation used throughout the remainder of this paper.

B. Series DC Connection and Power Flow

Neglecting auxiliary and bypass paths, the cell dc currents are constrained to be equal,

$$i_{dc,1} = i_{dc,2} = \dots = i_{dc,N} = i_{dc}, \quad (2)$$

so that the dc-side power entering cell k is

$$p_{dc,k} = V_{dc,k} i_{dc}. \quad (3)$$

The total machine power, by contrast, is the sum of the powers delivered to the individual winding groups,

$$P_{ac} = \sum_{k=1}^N P_k. \quad (4)$$

Under symmetrical operation $P_k \approx P_{ac}/N$, but equal power sharing is not a fundamental requirement of the topology. The multi-three-phase machine provides additional current and power degrees of freedom that allow power to be redistributed

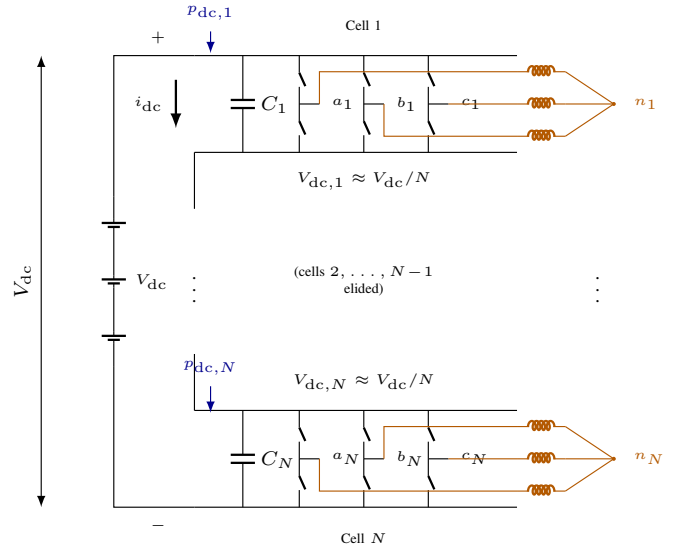


Fig. 5. Switch-level realization of the SPB dc-ac stage for two representative cells (Cell 1 and Cell N ; cells $2, \dots, N-1$ elided). The common series current i_{dc} (2) flows through every cell, giving each cell the dc-side power $p_{dc,k} = V_{dc,k} i_{dc}$ (3).

among winding groups [44], [54]. This redistribution capability is the physical basis for both dc-link voltage balancing and several of the post-fault operating strategies.

The combination of a common dc current with independently controllable ac-side powers also explains why the local capacitor voltages must be treated as internal dynamic states rather than fixed design parameters. If one winding group processes more power than another, the corresponding capacitor energies can diverge even though all cells carry the same dc current [45], [46]. Managing this divergence is the dc-link voltage-balancing problem, SPB's first-order control challenge.

C. Cell Count as a Converter–Machine Design Variable

Cell count is consequently a converter–machine design variable rather than a semiconductor variable alone. Increasing N also reduces the voltage available to each winding group and alters the number of winding sets, terminal count, electromagnetic coupling, current-control degrees of freedom, and available reconfiguration states.

Machine phase current, however, does not generally scale as $1/N$: it instead depends on the number of turns, flux linkage, winding connection, torque constant, and machine geometry. A many-cell architecture may therefore reduce the semiconductor voltage class while still requiring high-current devices or parallel chips at the device level – a coupling among cell count, voltage class, and current rating that matters most as the design space extends toward lower-voltage device classes.

IV. ELECTRIC-MACHINE REQUIREMENTS AND INTEGRATED IMPLEMENTATION

A. Winding Architecture and Electromagnetic Coupling

Although the winding groups are electrically separated at their terminals, they share the same magnetic circuit, rotor,

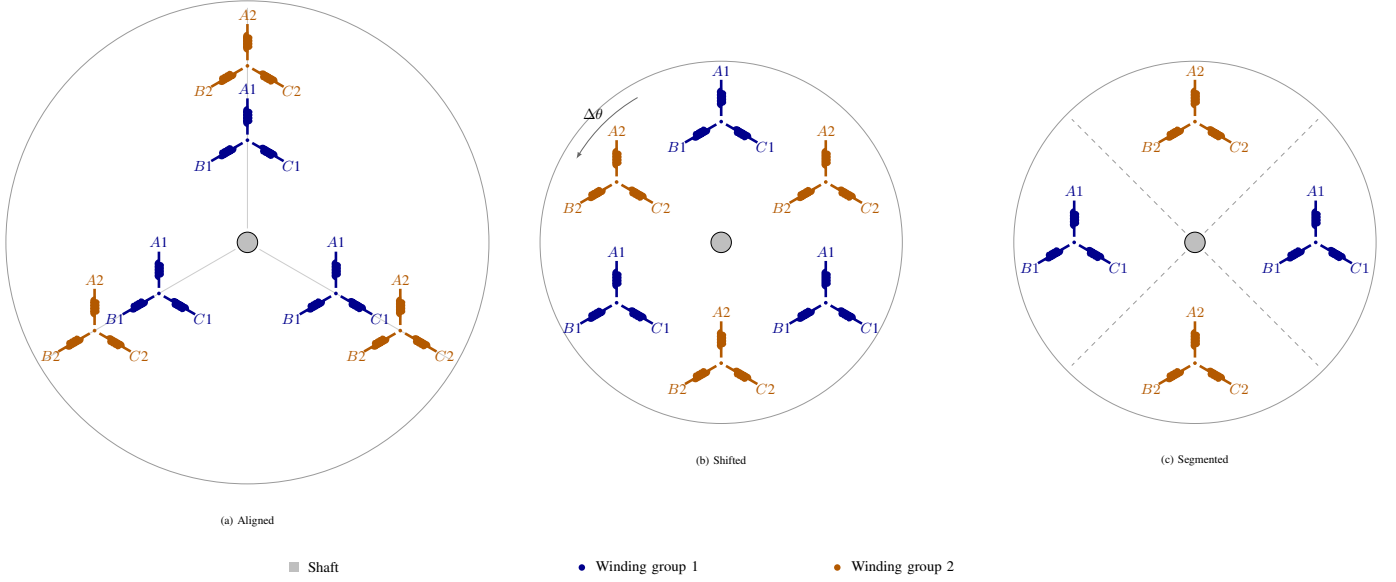


Fig. 6. Schematic comparison of the winding-group spatial arrangements : (a) aligned groups (b) spatially shifted groups and (c) physically segmented stator modules.

air gap, and shaft. Their electromagnetic coupling depends on winding distribution, slot/pole combination, spatial displacement, and machine geometry [33], [44], [54], [58]. Therefore, an SPB-compatible machine cannot generally be obtained simply by dividing a conventional three-phase winding into multiple groups without considering the resulting coupling and harmonic behavior.

The three-phase groups may be aligned, spatially shifted, or implemented as physically segmented stator modules. Spatial displacement affects mutual coupling, torque-ripple components, and the additional current subspaces available to the controller, among other effects [59]–[61]. These additional subspaces can introduce non-torque-producing currents, but they can also be exploited for harmonic suppression, differential power sharing, and fault-tolerant operation [54], [62]. Split winding architectures show a similar tradeoff: they can reduce ripple and provide additional current/voltage degrees of freedom, but they do so by increasing winding terminals and coupling complexity [63]. These three winding-group arrangements are compared schematically in Fig. 6.

For the k th PMSM winding group,

$$\begin{aligned} v_{d,k} &= R_{s,k} i_{d,k} - \omega_e L_{q,k} i_{q,k}, \\ v_{q,k} &= R_{s,k} i_{q,k} + \omega_e (L_{d,k} i_{d,k} + \psi_{f,k}), \end{aligned} \quad (5)$$

and the required voltage must satisfy

$$\sqrt{v_{d,k}^2 + v_{q,k}^2} \leq k_{\text{mod}} V_{\text{dc},k} \approx k_{\text{mod}} \frac{V_{\text{dc}}}{N}. \quad (6)$$

This falling per-cell voltage budget ties winding turns, inductance, and field-weakening capability directly to N – a coupling demonstrated in optimized multiphase SPB-PMSM systems [33]. Recent multiphase traction-machine studies similarly emphasize coordinated design of winding voltage, harmonic subspaces, and modulation [64], [65].

B. From Multiphase Machine to Integrated Modular Drive

Physical integration can convert the added electrical complexity of many winding groups into a packaging advantage. With a remote inverter, additional winding groups increase phase-cable and connector count; positioning each converter bridge close to its corresponding winding group instead reduces external cabling, shortens high-frequency interconnections, distributes heat sources, and aligns converter modules with stator segments.

This design philosophy predates SPB and has been extensively studied in integrated and modular motor drives [30], [31], [34], [38], [39], [66]–[78], summarized in Table II. The overall trend runs from early cable-reduction and compact-packaging concepts through jointly designed fault-tolerant converter/winding/thermal architectures and series-connected GaN IMMDS to recent WBG-based axial-flux IMMDS demonstrating stator-mounted converter modules and scalability to MW-class and EV-traction implementations.

C. Physical Integration Options

The electrically separated winding groups required by SPB do not impose a unique converter placement on a radial-flux machine. Chowdhury *et al.* [79], consistent with the broader integrated-motor-drive survey of Abebe *et al.* [80], classify motor/inverter physical integration into four categories: whether the converter mounts radially or axially, and whether it interfaces with a separate housing/end-cap structure or directly with the stator core. This taxonomy frames how SPB converter cells can be placed around a radial-flux machine, as illustrated in Fig. 7. The axial-flux stator-mounted and polygon-housing alternative is covered by the Mohamed *et al.* developments [74]–[76] in Table II.

Modern WBG drives therefore provide several machine and packaging concepts that are compatible with SPB, including

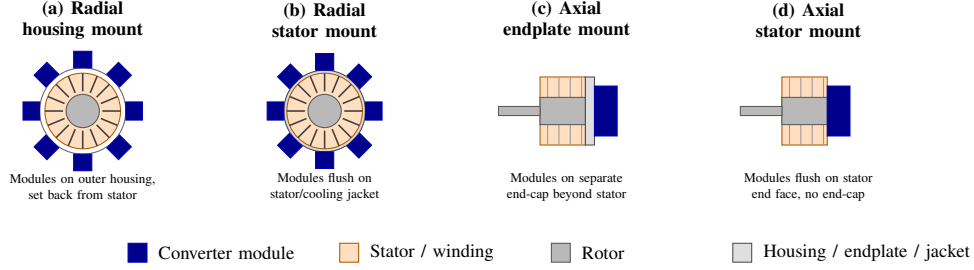


Fig. 7. Physical integration options for SPB-compatible converter modules on a radial-flux machine, following the four-category taxonomy of [79]: (a) radial housing mount, (b) radial stator mount, (c) axial endplate mount, and (d) axial stator mount, each shown in whichever radial or axial cross-section most clearly conveys its mounting geometry.

TABLE II
REPRESENTATIVE INTEGRATED MOTOR-DRIVE DEVELOPMENTS

Reference	Integration theme
Shakweh <i>et al.</i> [66]	Integrated motor/inverter architecture
Wheeler/Pickering <i>et al.</i> [67], [68]	Converter and thermal design inside motor envelope
Brown <i>et al.</i> [69]	Converter mounted at modular stator pole
Wolmarans <i>et al.</i> [70]	Fault-tolerant converter, machine, and thermal design
Choi <i>et al.</i> [30]	Modular PM machine and high-temperature electronics
Wang <i>et al.</i> [31], [38]	Low-voltage GaN, dc-voltage sharing, and interleaving
Uğur and Keysan [72], [73]	Multi-physics converter-machine optimization
Mohamed <i>et al.</i> [74]–[76]	Stator-mounted modules, polygon housing, integrated dc link
Wang <i>et al.</i> [77]	Repeated machine and converter submodules
Chen <i>et al.</i> [78]	WBG converter, machine, passive, and cooling co-design
Deng <i>et al.</i> [34]	Recent IMD technologies and research directions

segmented windings, distributed converter cells, shared cooling, and integrated dc-link structures. These developments do not establish an intrinsic SPB advantage, but they demonstrate the availability of enabling technologies required for practical implementation.

D. Integration Is Not Free of Complexity

Integration does not remove complexity; it relocates it. Converter modules mounted on or near the machine are exposed to elevated temperature, vibration, and sealing/coolant requirements, among other stresses, and to electromagnetic coupling between gate-drive, power, and machine conductors. Shared machine-converter cooling can also increase thermal interaction between the stator and power electronics.

SPB introduces an additional insulation challenge because its converter cells and associated winding groups occupy different absolute potentials along the series dc stack. Gate-driver supplies, sensors, and mechanical mounting, among other subsystems, must therefore preserve the required electrical isolation.

Consequently, practical SPB integration requires coordinated electrical, thermal, and mechanical design, complementing the semiconductor- and system-level tradeoffs.

TABLE III
SPB MODULATION DEGREES OF FREEDOM GROUPED BY PRINCIPAL SYSTEM-LEVEL EFFECT.

Effect	Degrees of freedom
Ripple / loss shaping	Carrier phase [47], [51], [81]; switching sequence [82], [83]; clamping/DPWM [84], [85]
CMV / balancing authority	Zero-sequence & vector redundancy [86], [87]; differential cell commands [54], [88]

V. MODULATION STRATEGIES

Each SPB cell is electrically simple while the resulting control problem is not. Cell dc current is shared, cell dc voltage is a dynamic state, and the winding groups are electromagnetically coupled. Modulation is the layer at which these converter- and machine-side constraints are jointly realized. Since each SPB cell is a conventional three-phase two-level bridge, standard SPWM, SVM, and DPWM methods can be applied locally without modification. The distinctive feature of SPB is that modulation can additionally be coordinated between cells through carrier phase, zero-sequence injection, switching sequence, and clamping intervals [43], [47]. These inter-cell degrees of freedom affect dc-link ripple, machine harmonics, semiconductor switching loss, common-mode voltage (CMV), and the power exchanged by the individual winding groups. Table III groups these degrees of freedom by their principal system-level effect.

A. Carrier Coordination and Harmonic Cancellation

The switching function of phase leg x in cell k has the usual double-Fourier form,

$$\begin{aligned}
 s_{x,k}(t) = & \frac{A_{00}}{2} + \sum_{n=1}^{\infty} [A_{0n} \cos(n[\omega_{0,k}t + \theta_{0,k}]) + B_{0n} \sin(\cdot)] \\
 & + \sum_{m=1}^{\infty} [A_{m0} \cos(m[\omega_{c,k}t + \theta_{c,k}]) + B_{m0} \sin(\cdot)] \\
 & + \sum_{m=1}^{\infty} \sum_{\substack{n=-\infty \\ n \neq 0}}^{\infty} [A_{mn} \cos(m[\omega_{c,k}t + \theta_{c,k}] \\
 & \quad + n[\omega_{0,k}t + \theta_{0,k}]) + B_{mn} \sin(\cdot)], \quad (7)
 \end{aligned}$$

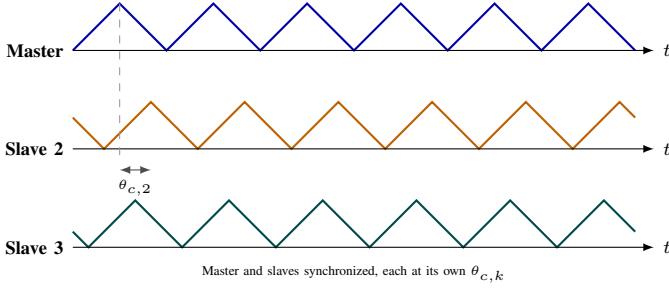


Fig. 8. Master/slave carrier synchronization for a representative $N = 3$ example: each slave's carrier is offset from the master by its own phase $\theta_{c,k}$ (8).

with carrier index m (at carrier frequency $\omega_{c,k}$, phase $\theta_{c,k}$) and sideband index n (at fundamental frequency $\omega_{0,k}$, phase $\theta_{0,k}$).

Each cell's carrier phase $\theta_{c,k}$ is a free parameter in this expansion, giving SPB a degree of freedom for reshaping the aggregate harmonic content. Uniform interleaving exploits this freedom by assigning each cell's carrier phase as

$$\phi_{c,k} = \phi_{c,1} + (k-1)\Delta\phi_c, \quad \Delta\phi_c = \frac{2\pi}{N}. \quad (8)$$

Fig. 8 illustrates this conceptually for a master cell and two slave cells.

B. DC-Link Ripple Reduction and Battery-Side Current

DC-link current ripple accelerates battery aging and is itself a design objective. Lower-voltage cell switches typically enable a higher switching frequency for a given loss budget, reducing the required per-cell capacitance and battery-side ripple relative to a single high-voltage bridge; carrier-phase interleaving reduces this ripple further still.

The cell's dc-side current follows from convolving each phase's switching-function spectrum with its current spectrum and summing over the three phases,

$$I_{\text{conv},x,k} = \frac{1}{2} (S_{x,k} \otimes I_{x,k}). \quad (9)$$

For a single cell, every carrier harmonic m contributes sidebands at $n = \pm 6p - 3$ (m odd) or $n = \pm 6p$ (m even), for integer p . With uniform interleaving $\theta_{c,k} = 2\pi k/N$, summing the N cells' converter currents shows that only carrier harmonics that are themselves multiples of N survive ($m = pN$ for even N , $m = 2pN$ for odd N), each retaining the same sideband structure [43].

Translating this converter-current spectrum into the dc-link current actually shared by the series stack requires the SPB equivalent circuit of Section VII,

$$L \frac{di_{\text{dc}}}{dt} = E - Ri_{\text{dc}} - \sum_k v_{\text{conv},k}, \quad C_k \frac{dv_{\text{conv},k}}{dt} = i_{\text{dc}} - i_{\text{conv},k}, \quad \text{Loss} \quad (10)$$

giving a closed-form transfer function between the summed converter-current spectrum and the resulting dc-bus current [43],

$$I_{\text{dc}}(\omega) = \frac{\sum_{k=1}^N I_{\text{conv},k}(\omega)}{N - \omega^2 LC + j\omega CR}. \quad (11)$$

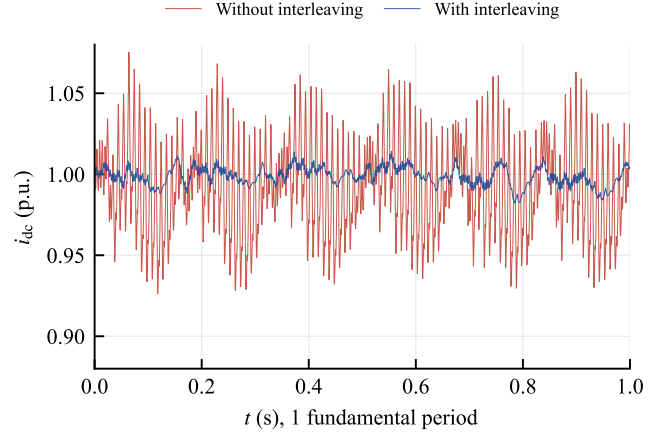


Fig. 9. Coupled circuit simulation of the shared series dc-link current $i_{\text{dc}}(t)$ for $N = 3$ SPB cells ($M = 0.85$, $f_1 = 1$ Hz, $f_{sw} = 100$ Hz), without (red) and with (blue) carrier interleaving between submodules (8). Both traces share the same dc average; interleaving reduces the peak-to-peak ripple by roughly 4–5 $\times$, consistent with (12). Computed simulation result, not measured hardware data.

Because the residual dc-link ripple is dominated by the lowest surviving harmonic, $m = N$, evaluating (11) there gives an approximate ripple estimate,

$$\Delta i_{\text{dc}} \approx \frac{\left| \sum_{k=1}^N I_{\text{conv},k}(2\pi N f_{sw}) \right|}{\left| N - (2\pi N f_{sw})^2 LC + j2\pi N f_{sw} CR \right|}. \quad (12)$$

Because the surviving harmonic sits at N times the per-cell switching frequency, both the removal of the uncanceled low-order sidebands and the higher-frequency LC attenuation act in the same direction. Fig. 9 illustrates this by simulation for a representative $N = 3$ case.

This simulation lumps the battery-to-stack bus inductance but treats each cell's own connection to that bus as ideal. In a physically distributed implementation, each cell instead reaches the shared dc path through its own bus-bar and connector loop, whose parasitic inductance couples back into the cancellation mechanism itself: Wang *et al.* [39] found that a series-connected GaN IMMD required careful capacitor selection and tight layout to preserve the benefit of interleaving, and an earlier characterization of an interleaved series-connected drive found that parasitic loop inductance could make the interleaved ripple larger than the non-interleaved case [89]. The reduction in Fig. 9 should therefore be read as an upper bound, with the achievable benefit set by layout, capacitor selection, and switching frequency rather than by carrier phasing alone.

C. Machine Harmonics, Voltage Utilization, and Switching

The optimum carrier displacement is nonetheless not universal: it depends on cell count, modulation index, power factor, winding displacement, and the specific quantity being minimized, so carrier coordination should be viewed as one design variable within a larger converter-machine modulation problem.

SPB-specific studies show that inter-cell modulation affects both semiconductor losses and dc-link-current harmonics [47], and related modular and multiphase drives demonstrate simultaneous reduction of dc-link and machine current ripple [51], [82], [90].

The multiphase machine makes the harmonic problem substantially more involved than in a conventional three-phase drive: nonfundamental voltage components can excite additional harmonic subspaces with low impedance, producing copper loss and torque ripple even when fundamental current control is fully satisfactory. Recent integrated-modular-drive studies accordingly move toward interleaved and discontinuous PWM techniques tailored to this multiphase current-ripple problem [81], [91], [92].

At high speed, modulation must also maximize voltage utilization, because each winding group is supplied only from its local cell voltage $V_{dc,k}$, as constrained by (6) – an objective that grows more important as N increases and $V_{dc,k}$ falls accordingly. Optimized vector selection for dual-three-phase traction machines has demonstrated the possibility of extending the high-modulation region while simultaneously suppressing unwanted subspace currents [64].

Switching loss introduces an opposing objective. DPWM and hybrid strategies reduce the number of transitions by clamping selected phases or by redistributing high-frequency switching among converter modules. Related open-end-winding (OEW) work demonstrates that coordinated modulation can reduce switching loss and unwanted zero-sequence current simultaneously [84], [85]; although its dc architecture differs, the underlying principle of distributing switching activity among several inverter modules is directly transferable to SPB.

D. Coupling to Balancing and Common-Mode Objectives

Modulation also sets the achievable authority for two further objectives that are developed as dedicated topics later in this paper: dc-link voltage balancing (Section VII), where differential winding-group power must be realizable through modulation without violating local voltage and current limits [54], [55], [88]; and common-mode voltage (Section VI), where redundant voltage vectors or coordinated zero-sequence terms can reduce frame excitation [86], [87], [93]. The key SPB modulation question is therefore not which conventional PWM method performs best in isolation, but how switching decisions should be coordinated across cells to jointly manage ripple, loss, balancing, and common-mode objectives over the full operating map.

VI. COMMON-MODE VOLTAGE EFFECTS

In any converter-fed ac machine, the non-zero average of the three phase-leg pole voltages produced by PWM switching gives rise to a common-mode voltage (CMV) that excites parasitic capacitive paths between winding, frame, shaft, and bearings. This mechanism is well characterized for a converter with a single machine port whose voltage scaling is produced internally, such as the flying-capacitor multilevel converter, for which an analytical CM-EMI model referenced to one

dc-link potential exists [94]. SPB does not fit this picture: because its N cells are connected in series on the dc side (Section III) while each feeds an electrically separate winding group, cell k 's local dc rails are displaced from the overall dc-bus reference by an amount that depends on its position in the stack.

A. Series-Stack Origin of Common-Mode Excitation

Taking cell 1's negative dc rail as the reference, with cell k 's negative rail coinciding with cell $(k-1)$'s positive rail, the absolute offset of cell k is

$$V_{-,k} = \sum_{j=1}^{k-1} V_{dc,j} \approx (k-1) \frac{V_{dc}}{N}, \quad (13)$$

using the voltage-partitioning relation (1). The common-mode potential seen by winding group k is then

$$v_{cm,k}^{abs}(t) = V_{-,k} + \frac{1}{3} (v_{aN_k} + v_{bN_k} + v_{cN_k}), \quad (14)$$

where $v_{aN_k}, v_{bN_k}, v_{cN_k}$ are cell k 's pole voltages referenced to its own negative rail N_k . The second term is the conventional switching-induced CMV already addressed by zero-sequence injection and vector-redundancy modulation (Section V); the first term, $V_{-,k}$, is specific to SPB, set by the cell's stack position and by the instantaneous voltage-sharing state (Section VII), not by modulation alone.

Fig. 10 illustrates the resulting structure: increasing absolute potential offset with stack position, together with representative CM current paths coupling each winding group to the shared frame and chassis ground through parasitic capacitance. Because the cells and their winding groups are physically close together in the integrated implementations of Section IV, these paths are short and the coupling between cells is direct rather than mediated by long cabling. Physically integrated motor-drive assemblies show that common-mode current can resonate through exactly this kind of parasitic loop, and that the resonance is not adequately predicted by a single-source CM model [93], [95] – extending such modeling to a multi-port structure with N distinct dc offsets is, to the authors' knowledge, not yet available for SPB specifically.

B. Carrier Synchronization for CMV Cancellation

Beyond the stack-offset mechanism above, synchronizing each cell's PWM waveform relative to the others is a way to actively cancel the switching-induced component of CMV. We illustrate the principle for the $N = 2$ case, where the cancellation condition has a simple closed form; the same synchronization idea extends to larger stacks built from matched cell pairs, though a general N -cell synchronization law has not yet been derived for SPB.

If cell 2 is driven with the logical inverse of cell 1's gate commands, and cell 2's winding set is wired with reversed winding sense so that both cells still generate torque in the same direction, the two cells' individually switched

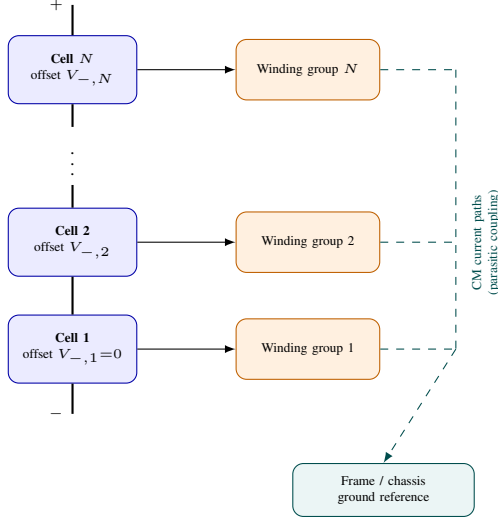


Fig. 10. Simplified SPB series-stack structure: absolute potential offset per cell (13) and representative CM current paths (dashed) to the shared frame/chassis ground through parasitic capacitance.

CMVs become approximately equal and opposite, $v_{cm,1}(t) \approx -v_{cm,2}(t)$, and the total CMV seen by the machine,

$$v_{cm,tot}(t) = \frac{v_{cm,1}(t) + v_{cm,2}(t)}{2}, \quad (15)$$

cancels toward zero instead of adding constructively. This balanced-cancellation principle was first demonstrated for a dual-winding-set machine driven by a balanced inverter pair on a single, non-series-stacked machine [96], and was later applied to a genuine two-cell series-stacked (2L-SSC) SPB inverter [57]; as with interleaved GaN IMMDs, the achievable reduction is ultimately limited by layout parasitics and local dc-link capacitor design [39].

This cancellation assumes perfectly matched, simultaneous switching transitions between the two cells: a propagation-delay mismatch (e.g., from PCB layout or driver differences) reintroduces a residual CM spike, of duration equal to the delay, at every misaligned transition, and unequal power sharing during dc-link balancing similarly degrades the cancellation, both shown quantitatively in [57], [96]. The mechanism is specific to $N = 2$, or to a stack of matched cell pairs, so it should be read as a valuable special-case mitigation rather than a general solution to the stack-offset problem above. Fig. 11 simulates this cancellation directly from switching-function waveforms, including its effect on the resulting bearing/ground current.

C. Physical Consequences and Mitigation

CMV-driven bearing current is a documented failure mechanism in inverter-fed machines generally, arising from capacitive coupling through the air gap and discharge events once the bearing lubricant film breaks down [97], and can be identified with dedicated non-intrusive current-measurement techniques [98]. In SPB, this mechanism acts at each of the N winding groups individually; because (13)–(14) gives each group a different absolute CM reference, the resulting bearing-

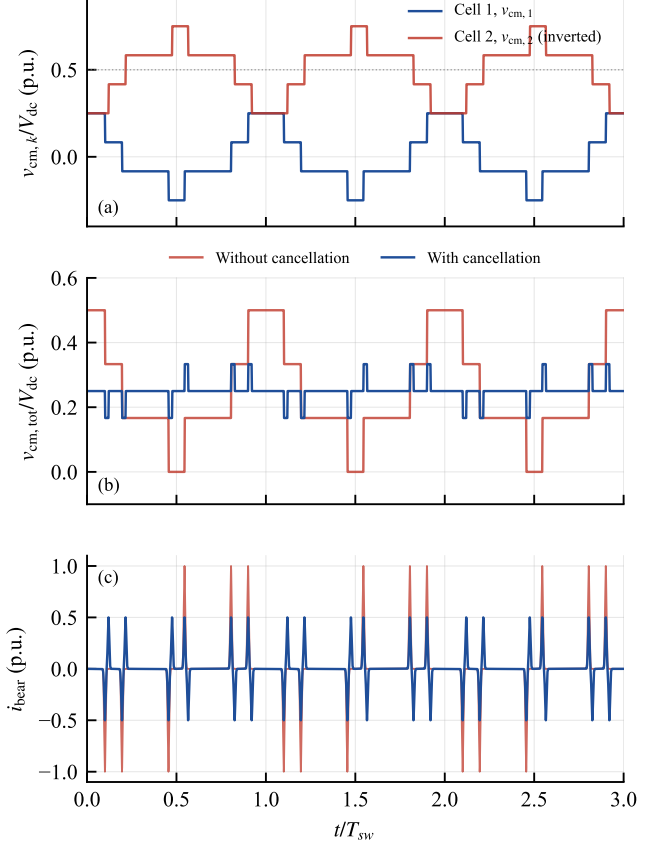


Fig. 11. Switching-function simulation of $N = 2$ SPB common-mode cancellation ($M = 0.85$, $\cos \varphi \approx 0.94$), all quantities in per unit, with cell 2 driven by the logically inverted gate commands of cell 1 plus a propagation delay $t_d = 0.02 T_{sw}$. (a) Each cell's absolute CM voltage $v_{cm,k}^{abs}$, combining the stack offset $V_{-,k}$ (13) with its local switching term. (b) Total CM voltage $v_{cm,tot}(t)$ (15): cancellation removes the switching ripple, leaving only delay-induced residual spikes on the same stack-offset average. (c) Resulting bearing/ground current $i_{bear} = C dv_{cm,tot}/dt$, normalized to the without-cancellation peak. Computed simulation result, not measured hardware data.

and shaft-voltage stress is not necessarily uniform across the stack, and whether it is dominated by the highest-offset cell, by all cells combined, or by resonant interaction between cells remains open. Shaft-voltage cancellation circuits developed for conventional drives [99] are a plausible mitigation candidate but have not been demonstrated for a series-stacked, multi-port machine.

The same switching events also produce fast dv/dt transitions at the winding terminals, stressing turn-to-turn and phase-to-frame insulation independently of the CM current path. WBG devices, already motivated for SPB by their favorable scaling at reduced blocking voltage (Section VIII), shorten rise time and increase overshoot, and have been shown to accelerate insulation aging and partial-discharge activity in hairpin and other traction-machine windings [100]–[102]; diagnostic methods exist to assess the resulting insulation condition [103], within the broader set of insulation technologies for transport-electrification machines [104]. In SPB, this stress is superimposed on the stack-position-dependent offset of (13), so insulation coordination must account for

both the local switching transition and the cell's absolute stack position, consistent with the challenge identified qualitatively in Section IV.

Mitigation therefore has to act at more than one level. Active EMI filtering can attenuate common-mode current at the converter terminals without relying on machine or bearing hardware [105], and cancellation circuits can target shaft voltage directly [99]. At the modulation level, Section V showed that zero-sequence injection and redundant-vector selection reduce CMV excitation [86], [87], but the carrier-phase displacement used there to interleave switching harmonics for ripple reduction, (8), is chosen to cancel a selected ripple harmonic and is not simultaneously optimized for common-mode excitation. Because SPB's CM problem additionally depends on the fixed stack offset $V_{-,k}$, which carrier phasing cannot change, an interleaving choice that minimizes dc-link or stator ripple need not minimize aggregate common-mode excitation, and vice versa – a genuine SPB-specific design tension that future studies should address by extending the multi-objective modulation problem of Section V to common-mode and insulation-stress metrics evaluated across the full stack.

VII. DC-LINK DYNAMICS, STABILITY, AND VOLTAGE BALANCING

The shared dc current and independently controllable winding-group powers (Section III) make the local capacitors coupled energy-storage states whose voltages are not guaranteed to remain equal [45], [46], a conclusion also reached independently for a series-connected multilevel motor drive, where distributed closed-loop current control alone was shown to leave the module voltages inherently unstable without dedicated balancing action [89].

For cell k ,

$$C_k V_{dc,k} \frac{dV_{dc,k}}{dt} = V_{dc,k} i_{dc} - p_{ac,k} - p_{loss,k}. \quad (16)$$

This captures the essential balancing mechanism: dc input power depends directly on local cell voltage because the series current is common, while output power is set independently by the corresponding winding group's controller.

For analysis, define

$$\tilde{V}_k = V_{dc,k} - \frac{1}{N} \sum_{j=1}^N V_{dc,j}. \quad (17)$$

A simplified differential-mode model around a balanced point is

$$CV_0 \frac{d\tilde{V}_k}{dt} \approx I_{dc,0} \tilde{V}_k - \Delta P_k. \quad (18)$$

In motoring operation ($I_{dc,0} > 0$), a positive cell-voltage perturbation increases that cell's dc input power and can reinforce the deviation if the ac-side power is unchanged; in regeneration the feedback sign reverses. This simplified picture is consistent with the detailed SPB stability analyses in [45], [46] and with recent work on related series-connected multiphase drives [106].

Reference [46] derives this instability directly from the full nonlinear constant-power-load model, without the small-signal

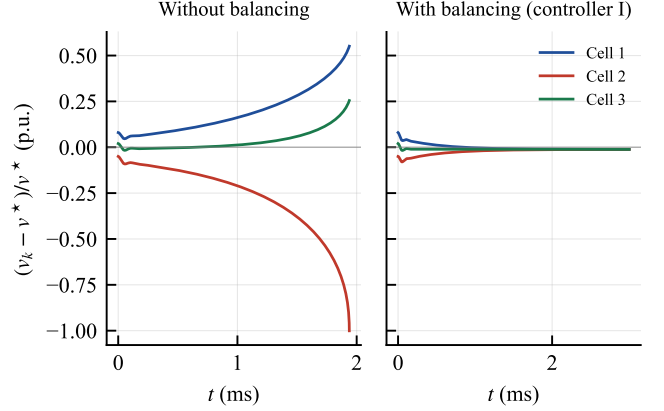


Fig. 12. Nonlinear dc-link voltage-sharing simulation for $N = 3$ cells, without (left) and with (right) the balancing controller of [46]. Computed simulation result, not measured hardware data.

approximation of (18): with $L_b di_{dc}/dt = E_b - R_b i_{dc} - \sum_k V_{dc,k}$ and $C_k dV_{dc,k}/dt = i_{dc} - P_k/V_{dc,k}$, the open-loop characteristic polynomial has $N-1$ real, strictly positive eigenvalues in motoring operation ($P_k > 0$) for any $N > 1$ – the series stack is provably unstable without a balancing controller, regardless of parameter choice. Fig. 12 integrates this exact nonlinear model using the real experimental parameters of [46] ($L_b = 8 \mu\text{H}$, $R_b = 0.35 \Omega$, $C = 100 \mu\text{F}$ per cell): without balancing, a small initial imbalance collapses one cell's voltage within milliseconds, while the proportional balancing law of [46] (their “controller alternative I”) drives the same imbalance back to zero on a comparable timescale.

The multiple winding groups provide a natural actuator for controlling internal capacitor energy. The cell power reference can be decomposed as

$$P_k^* = \frac{P^*}{N} + \Delta P_k, \quad \sum_{k=1}^N \Delta P_k = 0, \quad (19)$$

so differential power redistributes internal energy without changing the requested total machine power. In practice ΔP_k is limited by phase-current margin, winding voltage, torque-ripple constraints, and thermal state, so balancing authority is operating-point dependent.

Three main SPB balancing approaches can be identified. Centralized active balancing uses measured capacitor voltages to generate differential current or power commands [45], [46]. Common-duty-ratio (CDR) control applies common modulation commands and can obtain stable voltage sharing from the natural converter-machine dynamics for the investigated induction-machine system [52], though practical timing mismatch remains important [55]. Distributed multi-agent control uses local measurements and neighbour communication, and has been demonstrated for voltage balancing, power sharing, and reconfiguration [53], [54], [56], comparable to the local droop laws that coordinate power sharing across modules in the speed-droop framework of Galassini *et al.* [107]. Table IV summarizes these approaches and their principal considerations.

TABLE IV
PRINCIPAL SPB DC-LINK BALANCING APPROACHES AND THEIR MAIN CONSIDERATIONS.

Approach	Mechanism	Key consideration
Centralized	Differential current/power commands [45], [46]	Direct authority; global sensing and coordination
CDR	Common modulation, natural coupled dynamics [52], [55]	Low overhead; symmetry- and timing-sensitive
Distributed	Local agents, neighbour communication [53], [54], [56]	Scalable; requires communication and synchronization

For automotive operation, balancing must remain effective through torque transients, field weakening, parameter mismatch, and module reconfiguration: voltage-sensor offset, communication latency, propagation delay, and the available differential current/power margin can determine the usable cell-voltage margin and, consequently, the semiconductor rating.

This underscores the section's central point: dc-link stability belongs to the complete converter-machine-control system, not the power stage alone.

VIII. SEMICONDUCTOR TECHNOLOGY AND CELL-COUNT SCALING

Semiconductor voltage scaling (Section I-B) and the voltage-versus-component-count tradeoff established in Section III are quantified here at the cell-count level. In an N -cell SPB, the dc-link voltage is distributed among the series-connected converter cells, allowing the use of lower-voltage semiconductor classes as N increases.

The device rating must be selected from the maximum expected blocking stress, not the nominal voltage division alone:

$$V_{\text{rated}} \geq k_{\text{rel}} \left(\frac{V_{\text{dc,max}}}{N} + \Delta V_{\text{imb}}^{\text{max}} + \Delta V_{\text{sw}}^{\text{max}} \right), \quad (20)$$

where $\Delta V_{\text{imb}}^{\text{max}}$ denotes the maximum cell-voltage imbalance, $\Delta V_{\text{sw}}^{\text{max}}$ the commutation overshoot, and k_{rel} the selected reliability margin. Cell count is therefore linked to discrete commercial voltage classes rather than to a continuous semiconductor-scaling law.

As introduced in Section I-B, SiC currently provides the most direct route for high-voltage traction, whereas GaN becomes increasingly attractive once the architecture reduces the per-device voltage enough to access lower-voltage classes with favorable conduction and switching characteristics [24], [35], [57].

The resulting discrete design transitions are summarized and visualized directly in Fig. 13. For a 1200-V dc link, $N = 2$ gives a nominal cell voltage of 600 V and therefore leaves limited transient margin for a 650-V device. Increasing the cell count to $N = 3$ reduces the nominal cell voltage to 400 V and provides a substantially more practical operating region for the 650-V class. Further increases in N give access to progressively lower-voltage device families, as labeled directly on Fig. 13. The potential semiconductor benefit is therefore

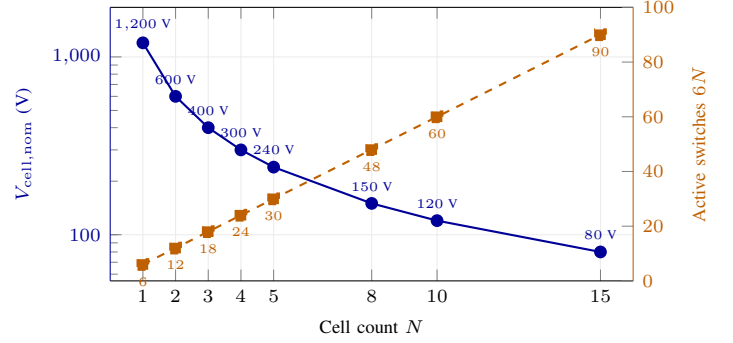


Fig. 13. Cell-count scaling for a 1200-V dc link: nominal cell voltage $V_{\text{cell,nom}} = V_{\text{dc}}/N$ (solid markers, left log axis) and active-switch count $6N$ (dashed markers, right linear axis) versus cell count N ; every marker is labeled directly with its exact value ($N = 1$: 1200 V, 6; $N = 2$: 600 V, 12; $N = 3$: 400 V, 18; $N = 4$: 300 V, 24; $N = 5$: 240 V, 30; $N = 8$: 150 V, 48; $N = 10$: 120 V, 60; $N = 15$: 80 V, 90). Increasing N opens progressively lower commercial semiconductor voltage classes (e.g., 1700–2000-V SiC at $N = 1$, 650-V GaN/SiC at $N = 3$, 100-V GaN/MOSFET class at $N = 15$) while the active-switch count grows linearly as $6N$.

nonlinear because a small change in N can open a new commercial voltage class, whereas the active-switch count grows approximately linearly, as plotted directly from these design points in Fig. 13.

A. Commercial-Device Screening

Access to a lower voltage class does not by itself imply a superior converter: the comparison must be based on commercial devices and their conduction, switching, current, thermal, and packaging constraints.

For nonlinear output capacitance, the stored energy

$$E_{\text{oss}}(V) = \int_0^V v C_{\text{oss}}(v) dv \quad (21)$$

is more informative than a single tabulated C_{oss} value. Depending on the commutation process, some or all of this stored energy can contribute to switching loss. Similarly, simple screening metrics such as $R_{\text{DS(on)}}C_{\text{oss}}$ or $R_{\text{DS(on)}}Q_g$ are useful for identifying candidate devices, but are insufficient for the final architecture comparison.

The principal device quantities required for SPB cell-count screening, and their primary role in the comparison, are: $R_{\text{DS(on)}}(T)$, which sets the die-area or parallel-device requirement; $E_{\text{oss}}(V)$, which sets the large-signal switching penalty; $E_{\text{on/off}}(V, I, T)$, which forms the operating-point switching-loss model; gate charge Q_g and Q_{gd} , which govern driver sizing and auxiliary loss; continuous current capability I_{cont} and thermal resistance R_{θ} , which govern parallelization and cooling requirements; and package or loop inductance L_{σ} , which sets the usable blocking-voltage margin against di/dt -induced overshoot.

Current capability is particularly important because machine phase current does not generally decrease in proportion to $1/N$ (Section III). If the derated current capability of one device is $I_{\text{dev,der}}$, the required number of parallel devices can be estimated as

$$n_p = \left\lceil \frac{I_{\text{sw,max}}}{I_{\text{dev,der}}} \right\rceil, \quad N_{\text{sw,tot}} = 6Nn_p. \quad (22)$$

Thus, a lower-voltage device can offer an attractive device-level FOM while still requiring substantial die area, package count, and gate-drive/thermal interfaces. Recent high-current 650-V GaN modules illustrate the importance of low-inductance packaging and integrated gate driving when GaN is scaled toward traction-current levels [108], [109].

Fast switching further couples device selection to layout. The commutation overshoot can be approximated by

$$\Delta V_{sw} \approx L_{\sigma} \frac{di}{dt}, \quad (23)$$

so the benefit of reduced nominal cell voltage can be lost if package and interconnect inductance consume a substantial fraction of the available blocking margin. This is particularly relevant for GaN, where high di/dt places strong requirements on the commutation loop, local decoupling, and gate-drive layout [109].

GaN device selection also requires consideration of dynamic $R_{DS(on)}$, reverse conduction, gate reliability, short-circuit capability, and thermal performance [110], [111]. These effects do not negate the benefit of lower blocking voltage, but they prevent room-temperature static $R_{DS(on)}$ from being used as the sole criterion for cell-count selection.

B. Loss Scaling With Cell Count

For one device, the dominant semiconductor losses can be represented approximately as

$$P_{cond} \approx I_{rms,sw}^2 R_{DS(on)}(T_j), \quad (24)$$

and

$$P_{sw} \approx f_{sw} [E_{on}(V, I, T_j) + E_{off}(V, I, T_j)]. \quad (25)$$

At converter level,

$$P_{semi}(N) = \sum_{k=1}^N \sum_{j=1}^{6n_p(N)} (P_{cond,k,j} + P_{sw,k,j} + P_{dt,k,j}) + P_{gate}(N). \quad (26)$$

The dependence on N is therefore not linear. The number of active devices increases with cell count, whereas $R_{DS(on)}$, switching energy, thermal impedance, and parallel-device requirement can change abruptly when the design enters a different semiconductor voltage class.

Consequently, SiC and GaN should be regarded as different regions of the SPB design space rather than as a fixed technology ranking. Low-cell-count architectures can favor high-voltage SiC because of its mature voltage and current capability, whereas increasing N can make lower-voltage GaN attractive if the improvement in conduction, switching, packaging, or passive-component performance compensates for the additional converter modules.

Resolving the cell-count question therefore requires reading Fig. 13 jointly with the screening quantities above and the machine requirements of Section IV, rather than from any single figure of merit.

IX. FAULT TOLERANCE AND RECONFIGURATION

SPB modularity provides structural redundancy because the converter and machine are divided into repeated converter-winding modules. However, fault tolerance is not automatic: because the dc ports are connected in series, a failed cell can disturb or interrupt the complete dc current path, a structural challenge demonstrated for SPB and for other series-connected modular converters alike [112]–[114]. Early SPB work demonstrated post-fault operation under semiconductor faults, establishing that repeated hardware must be accompanied by appropriate safe states and control action [112]. At the machine level, a submodule short-circuit fault in a modular converter feeding fractional-slot concentrated-winding PM machines was shown to drive a fault current above rated current because of the strong NdFeB rotor magnetization, risking machine overheating; reducing rotor magnetization limits this fault current but also narrows the constant-power-speed range, a design tradeoff specific to fault-tolerant FSCW PM machines [113]. In the SPB machine designs of that study, post-fault short-circuit current with NdFeB magnets exceeds the healthy-phase current, whereas with ferrite magnets it stays below it, at the cost of becoming visibly non-sinusoidal from magnetic coupling with the adjacent healthy windings – a concrete illustration that PM material selection is itself a fault-tolerance design variable, not only a torque-density one. Fault handling in other series-connected modular converters further underscores that mitigation requires the control system to be explicitly designed for module isolation [114].

The machine provides a second layer of redundancy. Healthy winding groups can potentially redistribute current after phase or converter faults, as demonstrated in recent multiphase and modular-drive studies [62], [115], [116]. In SPB, however, this machine-side redundancy is useful only if the failed converter cell can be isolated or bypassed without breaking the series dc path.

Building on the online reconfiguration capability introduced in Section II [56], after a faulted module is isolated the remaining active set $\mathcal{A}$ must satisfy

$$P^* = \sum_{k \in \mathcal{A}} P_k^*, \quad (27)$$

with torque derating if current, voltage, or thermal limits prevent full redistribution. Reconfiguration can therefore provide both fault isolation and an operating-range degree of freedom because the number of active cells changes the voltage available to the remaining winding groups. Isolating a cell also raises the voltage that each surviving cell must share of the fixed dc-link voltage V_{dc} ((1), now divided among $|\mathcal{A}| < N$ active cells), which is an additional reason current or power derating in (27) may be needed after a fault, as illustrated in Fig. 15. Fig. 14 maps these fault locations onto the SPB structure.

Reliability remains ambiguous because increasing cell count provides more potential degraded states but also more components that can fail. System or mission reliability is therefore more meaningful than component count alone. Markov-chain analysis has been applied to related fault-tolerant IMMDS to distinguish healthy, degraded, and failed states [118], while

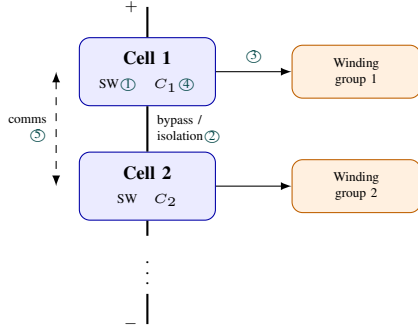


Fig. 14. Simplified SPB converter-machine structure with fault locations ①–⑤ annotated, corresponding in order to the rows of Table V.

TABLE V
SPB FAULT LOCATIONS AND REQUIRED SYSTEM RESPONSE

Fault	SPB consequence	Required response
Switch / bridge leg	Cell cannot synthesize commanded voltage	Fast safe state and protection [112]
Converter module	Series dc path can be interrupted	Bypass/isolation and power redistribution [56]
Winding / phase	Reduced and asymmetric torque capability	Multiphase post-fault current control [62], [116]
Capacitor / sensor*	Loss of voltage-sharing information or margin	Detection, derating, isolation or shutdown
Communication / controller	Loss of distributed coordination	Local fallback and deterministic safe state [117]

*No SPB-specific study directly reports capacitor or sensor fault handling; this row reflects the authors' system-level inference from the general modular-converter requirement to detect loss of cell-voltage sensing or local dc-link capacitance and to derate or isolate the affected cell accordingly.

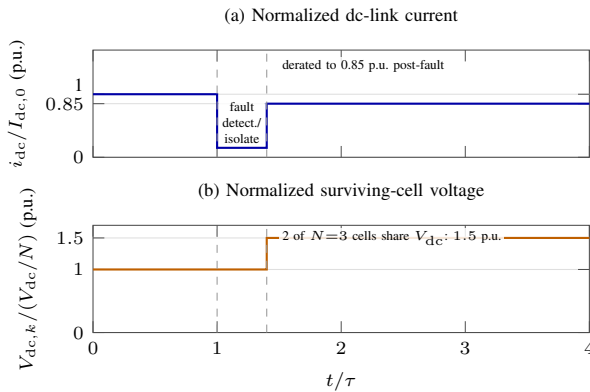


Fig. 15. Illustrative, analytically derived example of the post-fault current derating and remaining-cell voltage step-up for a representative $N = 3 \rightarrow 2$ case (one of three cells isolated after a fault); this is not measured or simulated data. (a) Normalized dc-link current, derated post-fault per (27). (b) Normalized remaining-cell voltage, stepping from 1 p.u. to $N/(N-1) = 1.5$ p.u. (1) once two cells must share V_{dc} . Both panels share the same time axis.

integrated distributed traction drives provide additional evidence that fault tolerance must include both the machine and the local converter architecture [70], [71].

The direct SPB literature therefore demonstrates the fea-

sibility of post-fault operation and online reconfiguration, but not yet automotive-level fault coverage. Detection and isolation time, short-circuit energy, capacitor and sensor faults, communication loss, post-fault thermal loading, diagnostic coverage, and deterministic functional-safety transitions remain important validation topics.

X. RESEARCH GAPS AND FUTURE DIRECTIONS

In theory, SPB is one of the more promising modular architectures for high-voltage electric traction: partitioning the dc-link voltage across series-connected cells grants access to lower-voltage semiconductor classes with a favorable figure of merit, supports modular converter-machine integration, and, as this review has shown, gives carrier-phase, balancing, and reconfiguration degrees of freedom that a conventional two- or three-level drive does not have. Whether these advantages translate into a real vehicle-level benefit, however, is not decided by any single subsystem but by how the battery/dc-link side, the semiconductor/converter side, and the machine side are co-designed together: the cell count that opens a favorable device voltage class also fixes winding voltage, current rating, and dc-link ripple frequency, and each of these choices feeds back into the others. This review has assembled the pieces needed for that co-design – voltage partitioning and cell-count scaling (Sections III and VIII), machine-integration options (Section IV), modulation and dc-link ripple (Section V), common-mode behavior (Section VI), dc-link balancing stability (Section VII), and fault tolerance (Section IX) – but, like the literature it surveys, has largely treated them as separate design axes rather than jointly.

A. Toward an Integrated Co-Design Framework

Future work should combine the voltage-partitioning, modulation and ripple, dc-link balancing, common-mode, and fault-derating relations developed in this review into a single N - and technology-dependent framework, so that cell count, semiconductor class, and machine design can be evaluated jointly against vehicle-level energy, cost, and reliability targets rather than compared through separate, narrower studies. Several concrete open questions define what such a framework must resolve.

The optimum cell count N remains the central unresolved design question: as established in Section III, the voltage-versus-component-count tradeoff shifts the optimum as SiC, GaN, packaging, and cooling evolve [3], [24]. A common-platform comparison of a few-cell 650-V GaN SPB against a many-cell low-voltage GaN SPB, using real commercial devices, would be valuable; the GaN comparison in [57] is a useful starting point but should be extended to automotive power with current parallelization, machine voltage/speed constraints, dc-link capacitance, cooling, and mission energy.

Converter, machine, and thermal models, among others, should be co-optimized rather than designed sequentially. Existing phase-count studies show machine design changes with SPB cell number [33]; the missing step is a complete automotive co-design using vehicle-level energy, volume, mass, cost, and reliability as a joint objective.

Reliability is part of that objective: future studies should combine semiconductor, capacitor, sensor, gate-driver, communication, and winding faults (Section IX; [56], [112]) in one state-based, Markov-style framework as used for related fault-tolerant IMMDS [118], keeping *component failure*, *module isolation*, and *loss of vehicle function* distinct. An N -cell drive may survive with $N - 1$ active modules, but its torque–speed envelope, module thermal loading, and dc-voltage margin all change, so reliability data should pair with a post-fault capability model – not a series reliability block diagram – to compare availability, limp-home power, and mission completion probability against efficiency.

Cost and sustainability assessment should use the same N -dependent framing: increasing N can reduce per-switch die cost but tends to increase isolated gate supplies, sensing channels, and assembly operations, among others, almost linearly, while a repeated converter–winding module can gain from common tooling and automated assembly – a balance likely to differ between high-volume passenger vehicles and lower-volume heavy-duty systems. The same accounting should separate production, operation, maintenance, and end-of-life impact over one vehicle mission and service life, since modular repair and longer service life can offset the added production impact of extra components.

Overall, SPB research must move from demonstrating that the architecture operates toward demonstrating where it is better: whether it gives a measurable vehicle-level advantage over mature two- and three-level architectures once machine design, thermal management, EMC, reliability, manufacturing, and cost are included.

XI. CONCLUSION

This paper reviewed stacked polyphase bridge (SPB) converters – series-stacked two-level bridges each feeding an electrically separated winding group – as a candidate architecture for future high-voltage traction drives, partitioning the dc-link voltage while introducing converter–machine modularity, distributed control, and fault-management opportunities.

The review also shows that the semiconductor voltage advantage cannot be separated from the complete traction system. Increasing cell count adds converter, control, and insulation overhead, so SPB does not eliminate the complexity of high-voltage conversion but redistributes it between semiconductor technology, machine architecture, and control. WBG development makes this trade increasingly relevant, particularly where SPB can move a high-voltage traction system toward lower-voltage GaN devices; whether that approach is advantageous depends on whether the device-level gains compensate for the additional modular hardware and machine requirements.

The remaining step is not further isolated proof-of-concept studies but an integrated co-design across the battery/dc-link, semiconductor/converter, and machine sides, evaluated jointly against vehicle-level energy, cost, and reliability targets rather than compared through separate, narrower studies. SPB should ultimately be judged not as another multilevel converter, but as a modular traction-drive architecture that exchanges

semiconductor blocking-voltage stress for converter–machine modularity. Its competitiveness depends on whether that exchange produces a measurable vehicle-level benefit.

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